

AWS GLUE JOBS/TRIGGERS

APARTADO A

1.- Crea un bucket S3 con una carpeta dentro, por ejemplo, clima/espana

En aws, vamos a la sección de s3 y creamos el bucket

The screenshot shows the AWS S3 console interface. At the top, there's a search bar with 'S3' and a 'Preguntar a Amazon Q' button. Below this, the 'Servicios' (Services) section is visible, showing 'S3' as 'Almacenamiento escalable en la nube' (Scalable cloud storage). The 'Crear bucket' (Create bucket) button is highlighted in orange. Below the buttons, the 'Configuración general' (General configuration) section is shown. It includes the 'Región de AWS' (AWS Region) set to 'EE.UU. Este (Norte de Virginia) us-east-1'. Under 'Tipo de bucket' (Bucket type), 'Uso general' (General purpose) is selected, with a description: 'Recomendado para la mayoría de los casos de uso y patrones de acceso. Los buckets de uso general son del tipo de bucket de S3 original. Permiten una combinación de clases de almacenamiento que almacenan objetos de forma redundante en múltiples zonas de disponibilidad.' The 'Nombre del bucket' (Bucket name) field contains 'clima-bucket-pi'. Below this, there's a note: 'Los nombres de los buckets deben tener entre 3 y 63 caracteres y ser únicos dentro del espacio de nombres global. Los nombres de los buckets también deben empezar y terminar con una letra o un número. Los caracteres válidos son a-z, 0-9, puntos (.) y guiones (-). Más información'. At the bottom, there are 'Cancelar' (Cancel) and 'Crear bucket' (Create bucket) buttons.

En la lista de buckets, hacemos click en el bucket que acabamos de crear y elegimos “crear carpeta”

🔍 *Buscar buckets por nombre*

	Nombre ▲
☐	aws-logs-777558146599-us-east-1
☐	clima-espana-bucket
☐	consultasclima
☐	demo-bucket-pff
☐	hdf-bucket

Acciones ▼ **Crear carpeta** **⬆ Cargar**

ra que otras personas obtengan acceso a sus objetos, tendrá

Introducimos el nombre y creamos la carpeta

Carpeta

Nombre de la carpeta

Los nombres de las carpetas no pueden contener "/". [Consulte las reglas de nomenclatura](#)

Cifrado del lado del servidor Información

El cifrado del lado del servidor protege los datos en reposo.

☐ La siguiente configuración de cifrado se aplica únicamente al objeto de carpeta y no a los objetos de subcarpeta.

Cifrado del lado del servidor

☒ **No especificar una clave de cifrado**
La configuración del bucket para el cifrado predeterminado se utiliza para cifrar el objeto de carpeta al almacenarlo en Amazon S3.

☐ **Especificar una clave de cifrado**
La clave de cifrado especificada se utiliza para cifrar el objeto de carpeta antes de almacenarlo en Amazon S3.

⚠ Si la política del bucket requiere que los objetos se cifren con una clave de cifrado específica, deberá especificar la misma clave de cifrado al crear una carpeta. De lo contrario, se producirá un error al crear la carpeta.

Cancelar **Crear carpeta**

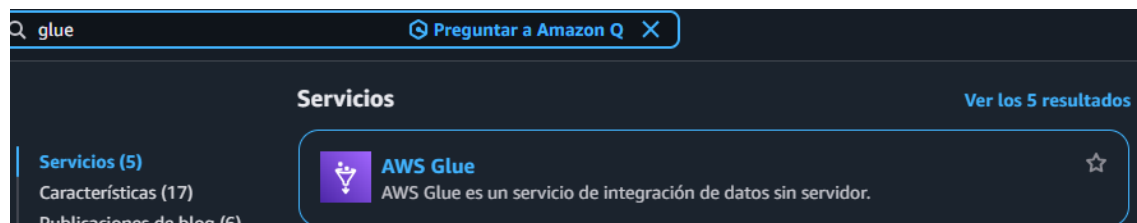
2.- Mediante un comando AWS CLI, copia los archivos csv con las mediciones de todas las estaciones metereológicas de España en él. Recuerda que la ruta era: s3://noaa-ghcn-pds/csv/by_station/). Y que los ID's (nombre de fichero, por tanto) de las estaciones de España comenzaba por "SP".

Utilizamos clli para copiar todos los archivos que empiezen por SP

```
PS C:\Users\Mañana> aws s3 cp s3://noaa-ghcn-pds/csv/by_station/ s3://clima-bucket-pr/espana/ --recursive --exclude "*" --include 'SP*'
copy: s3://noaa-ghcn-pds/csv/by_station/SP000004452.csv to s3://clima-bucket-pr/espana/SP000004452.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000003195.csv to s3://clima-bucket-pr/espana/SP000003195.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000008027.csv to s3://clima-bucket-pr/espana/SP000008027.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000008215.csv to s3://clima-bucket-pr/espana/SP000008215.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000008280.csv to s3://clima-bucket-pr/espana/SP000008280.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000008202.csv to s3://clima-bucket-pr/espana/SP000008202.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000008410.csv to s3://clima-bucket-pr/espana/SP000008410.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000006155.csv to s3://clima-bucket-pr/espana/SP000006155.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000008181.csv to s3://clima-bucket-pr/espana/SP000008181.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000008416.csv to s3://clima-bucket-pr/espana/SP000008416.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000009434.csv to s3://clima-bucket-pr/espana/SP000009434.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000060010.csv to s3://clima-bucket-pr/espana/SP000060010.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SP000009981.csv to s3://clima-bucket-pr/espana/SP000009981.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SPE00119711.csv to s3://clima-bucket-pr/espana/SPE00119711.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SPE00119729.csv to s3://clima-bucket-pr/espana/SPE00119729.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SPE00119738.csv to s3://clima-bucket-pr/espana/SPE00119738.csv
copy: s3://noaa-ghcn-pds/csv/by_station/SPE00119720.csv to s3://clima-bucket-pr/espana/SPE00119720.csv
```

3.- Crea una base de datos en el Data Catalog que se llame espana.

Entramos en AWS glue



Después en data catalog/databases y creamos la nueva base de datos

▼ Data Catalog

Databases

Tables

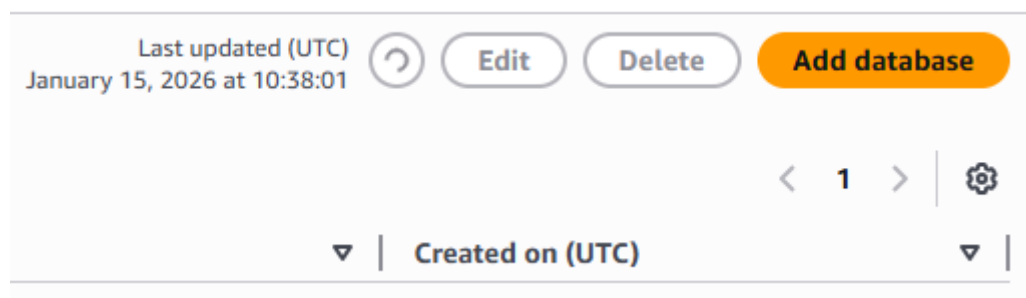
Stream schema registries

Schemas

Connections

Crawlers

Classifiers



Database details

Name

espana

Database name is required, in lowercase characters, and no longer than 255 characters.

Description - optional

Enter text

Descriptions can be up to 2048 characters long.

Database settings

Location - optional

Set the URI location for use by clients of the Data Catalog.

An S3 location is required for managed tables and Zero-ETL integrations.

Cancel
Create database

4.- Crea un Crawler que nos permita agregar a esa base de los ficheros de las estaciones meteorológicas de España (Pon como prefijo a la tabla espcsv_.

Entramos en Data Catalog/Crawlers

- ▼ Data Catalog
- Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

Y procedemos a crear el crawler haciendo click en “Crear Crawler” y siguiendo los pasos

Last updated (UTC)

January 15, 2026 at 10:55:20

Action ▼

Run

Create crawler

Set crawler properties

Crawler details info

Name

crawler-espana

Name can be up to 255 characters long. Some character set including control characters are prohibited.

Description - optional

Enter a description

Descriptions can be up to 2048 characters long.

► Tags - optional

Use tags to organize and identify your resources.

Cancel
Next

You don't have any data sources.

[Add a data source](#)

Add data source



Data source

Choose the source of data to be crawled.

S3



Network connection - optional

Optionally include a Network connection to use with this S3 target. Note that each crawler is limited to one Network connection so any other S3 targets will also use the same connection (or none, if left blank).



[Clear selection](#)

[Add new connection](#)

Location of S3 data

- ☒ In this account
- ☐ In a different account

S3 path

Browse for or enter an existing S3 path.



[View](#)

[Browse S3](#)

All folders and files contained in the S3 path are crawled. For example, type `s3://MyBucket/MyFolder/` to crawl all objects in MyFolder within MyBucket.

Subsequent crawler runs

This field is a global field that affects all S3 data sources.

- ☒ **Crawl all sub-folders**
Crawl all folders again with every subsequent crawl.
- ☐ **Crawl new sub-folders only**
Only Amazon S3 folders that were added since the last crawl will be crawled. If the schemas are compatible, new partitions will be added to existing tables.
- ☐ **Crawl based on events**
Rely on Amazon S3 events to control what folders to crawl.

☐ Sample only a subset of files

☐ Exclude files matching pattern

[Cancel](#)

[Add an S3 data source](#)

Choose data sources and classifiers

Data source configuration

Is your data already mapped to Glue tables?

☒ Not yet

Select one or more data sources to be crawled.

☐ Yes

Select existing tables from your Glue Data Catalog.

Data sources (1) [Info](#)

[Edit](#)

[Remove](#)

[Add a data source](#)

The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
☒ S3	s3://clima-bucket-pr/espana/	Recrawl all

► Custom classifiers - *optional*

A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.

[Cancel](#)

[Previous](#)

[Next](#)

Configure security settings

IAM role [Info](#)

Existing IAM role

LabRole



[View](#)

[Create new IAM role](#)

[Update chosen IAM role](#)

Only IAM roles created by the AWS Glue console and have the prefix "AWSGlueServiceRole-" can be updated.

Lake Formation configuration - *optional*

Allow the crawler to use Lake Formation credentials for crawling the data source. [Learn more](#).

☐ Use Lake Formation credentials for crawling S3 data source

Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, you must provide the registered account ID. Otherwise, the crawler will crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.

► Security configuration - *optional*

Enable at-rest encryption with a security configuration.

[Cancel](#)

[Previous](#)

[Next](#)

Set output and scheduling

Output configuration [Info](#)

Target database

espana



[Clear selection](#)

[Add database](#)

Table name prefix - *optional*

espcsv_

Maximum table threshold - *optional*

This field sets the maximum number of tables the crawler is allowed to generate. In the event that this number is surpassed, the crawl will fail with an error. If not set, the crawler will automatically generate the number of tables depending on the data schema.

Type a number greater than 0

► Advanced options

Crawler schedule

You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like [cron](#) syntax. [Learn more](#).

Frequency

On demand

[Cancel](#)

[Previous](#)

[Next](#)

5.- Guarda el Crawler pero no lo ejecutes.

Al final de los pasos, simplemente hacemos click en “Crear crawler”

Review and create

Step 1: Set crawler properties Edit

Set crawler properties

Name crawler-espana	Description -	Tags -
-------------------------------	-------------------------	------------------

Step 2: Choose data sources and classifiers Edit

Data sources (1) [Info](#)

The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
S3	s3://clima-bucket-pr	Recrawl all

Step 3: Configure security settings Edit

Configure security settings

IAM role LabRole	Security configuration -	Lake Formation configuration -
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Step 4: Set output and scheduling Edit

Set output and scheduling

Database espana	Table prefix - optional espcsv_	Maximum table threshold - optional -	Schedule On demand
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Cancel Previous Create crawler

APARTADO B

1. Crea una carpeta dentro del bucket anterior (clima) con el nombre parquet. por ejemplo, clima/parquet

Volvemos a la página de S3 y creamos una carpeta como en el apartado A ejercicio 1

Objetos (1) 🔄 Copiar URI de S3 Copiar URL ⬇ Descargar 🔗 Abrir 🗑 Eliminar Acciones ▾ Crear carpeta 📁 Cargar

Los objetos son las entidades fundamentales que se almacenan en Amazon S3. Puede utilizar el [inventario de Amazon S3](#) para obtener una lista de todos los objetos de su bucket. Para que otras personas obtengan acceso a sus objetos, tendrá que concederles permisos de forma explícita. [Más información](#)

☐

Nombre

▲

Tipo

▼

Última modificación

▼

Tamaño

▼

Clase de almacenamiento

▼

☐	 espana/		Carpeta	-	-	-	-
--------------------------	--	--	---------	---	---	---	---

Carpeta

Nombre de la carpeta

parquet/

Los nombres de las carpetas no pueden contener "/"-. Consulte las reglas de nomenclatura

Cifrado del lado del servidor

El cifrado del lado del servidor protege los datos en reposo.

La siguiente configuración de cifrado se aplica únicamente al objeto de carpeta y no a los objetos de subcarpeta.

Cifrado del lado del servidor

No especificar una clave de cifrado

La configuración del bucket para el cifrado predeterminado se utiliza para cifrar el objeto de carpeta al almacenarlo en Amazon S3.

Especificar una clave de cifrado

La clave de cifrado especificada se utiliza para cifrar el objeto de carpeta antes de almacenarlo en Amazon S3.

Si la política del bucket requiere que los objetos se cifren con una clave de cifrado específica, deberá especificar la misma clave de cifrado al crear una carpeta. De lo contrario, se producirá un error al crear la carpeta.

Cancelar

Crear carpeta

☐

Nombre

☐

espana/

☐

parquet/

2. Crea un trabajo mediante Visual ETL que nos permita cambiar el esquema de los CSV's que acabamos de importar poniendo los nombres de los campos en español y guardando los datos en formato parquet en la carpeta del punto anterior.

Buscamos AWS Glue y seleccionamos "AWS Glue Studio"

A screenshot of the AWS Glue Studio search results page. The search bar at the top contains the text "aws glue studio" and a button labeled "Preguntar a Amazon Q". Below the search bar, the results are categorized under "Servicios" (Services). The first result is "AWS Glue", which is highlighted. The description for AWS Glue is "AWS Glue es un servicio de integración de datos sin servidor." (AWS Glue is a serverless data integration service). Below the description, there are links for "Características principales" (Main features), "AWS Glue Studio", "Catálogo de datos" (Data catalog), "Rastreadores" (Tracers), "Flujos de trabajo" (Workflows), and "Set table properties".

Una vez dentro seleccionamos "Visual ETL"

AWS Glue Studio [Info](#)

Create job [Info](#)



Author in a visual interface focused on data flow.

Visual ETL

► Example jobs [Info](#)

En sources seleccionamos "Amazon s3"

+ Add nodes



Search sources, transforms and targets

Sources

Transforms

Targets

Popular



AWS Glue Data Catalog

AWS Glue Data Catalog table as the data source.



Amazon S3

JSON, CSV, or Parquet files stored in S3.



Amazon Kinesis

Read from an Amazon Kinesis Data Stream.



Apache Kafka

Read from an Apache Kafka or Amazon MSK topic.



Relational DB

AWS Glue Data Catalog table with a relational database as the data source.

Hacemos click en el s3 y seleccionamos el bucket que queremos utilizar



Data source - S3 bucket
Amazon S3



Name

Amazon S3

S3 source type

Info

☒ S3 location

Choose a file or folder in an S3 bucket.

☐ Data Catalog table

S3 URL

Q

s3://bucket/prefix/object

View

Browse S3

☒ Recursive

Read files in all subdirectories.

Data format

No format

✖

Must be a valid format.

Infer schema

► Additional options

Choose an S3 bucket

S3 buckets

Buckets (1/5)

Find bucket

Name

☐ aws-logs-777558146599-us-east-1

☒ clima-bucket-pr

☐ consultasclima

☐ demo-bucket-pff

☐ hdf-bucket

Cancel

Choose

Añadimos a la fuente la carpeta que queremos leer

S3 URL

Q

s3://clima-bucket-pr/espana/

✕

View

Browse S3

☒ Recursive

Read files in all subdirectories.

Data format

Elegimos formato CSV

Data source properties - S3

Name
Amazon S3

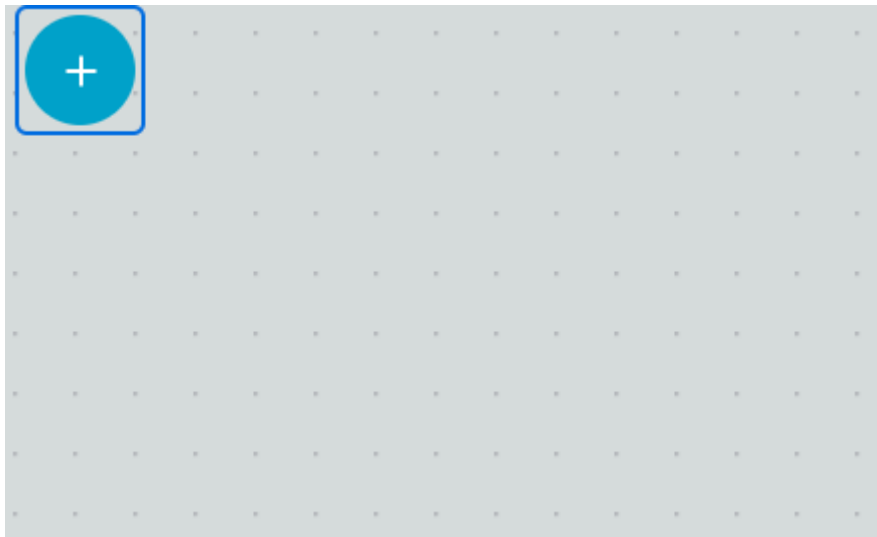
S3 source type | [Info](#)
☒ S3 location
Choose a file or folder in an S3 bucket.
☐ Data Catalog table

S3 URL
s3://clima-bucket-pr [View](#) [Browse S3](#)
☒ Recursive
Read files in all subdirectories.

Data format
CSV

Delimiter

Hacemos click en “Add notes” y después en “Transforms” y seleccionamos “Change Schema”



+ Add nodes

Q Search sources, transforms and targets

SourcesTransformsTargetsPopular



Data Preparation Recipe

Select and execute an external DataBrew Recipe.



Change Schema

Change field names, data types and drop fields. Formerly known as Apply Mapping.



Join

Combine records from two datasets based on a set of conditions.



SQL Query

Use a SQL query to transform data.



Detect Sensitive Data

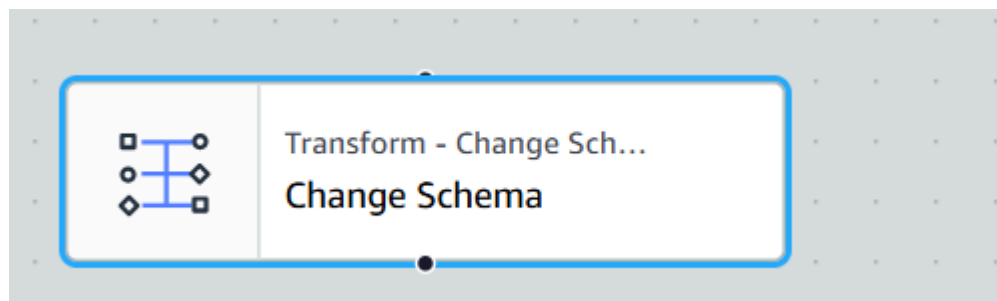
Detect PII and other sensitive information.



Evaluate Data Quality

Evaluate the quality and completeness of your data.

Hacemos click en “Change Schema” y en data source seleccionamos “Amazon S3”



Transform

Name

Change Schema

Node parents

Choose which nodes will provide inputs for this one.

Choose one or more parent node

Filter parent nodes

☒ Data sources

☒ Amazon S3
S3 - DataSource

☐ Transforms

☐ Unclassified nodes

Cambiamos los nombres de los campos que aparecen

Change Schema (Apply mapping)

Source key	Target key	Data type	Drop
ID	ID	string ▼	☐
DATE	FECHA	string ▼	☐
ELEMENT	ELEMENT	string ▼	☐
DATA_VALUE	VALOR DA	string ▼	☐
M_FLAG	BANDERA	string ▼	☐
Q_FLAG	BANDERA	string ▼	☐
S_FLAG	BANDERA	string ▼	☐
OBS_TIME	TIEMPO_C	string ▼	☐

Elegimos “LabRole” como el IAM Role en Data Preview

Start a data preview session

IAM role

To start a data preview session, choose an IAM role for this job. Changing the role will end an existing data preview session.

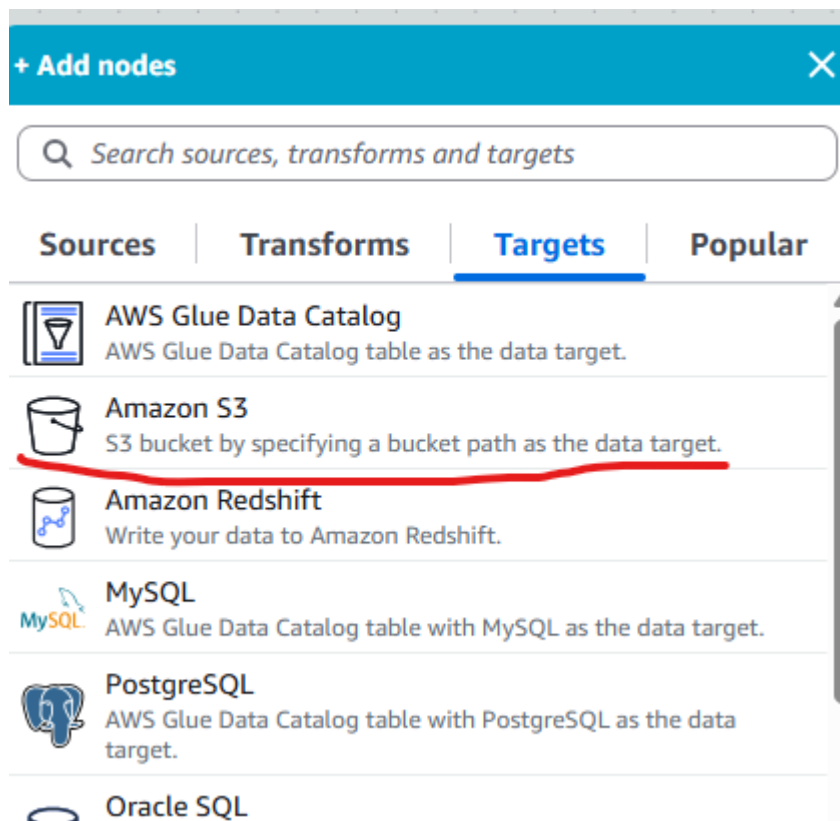
LabRole

No description available.

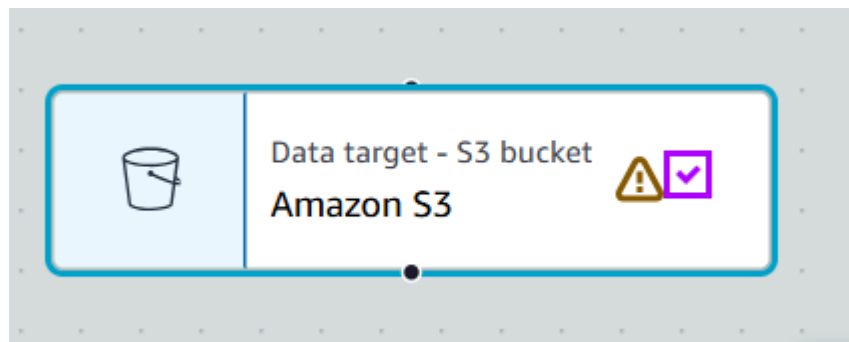
[Create IAM role.](#)

Additional Settings

Volvemos a “Add notes” y esta vez hacemos click en “Targets” y elegimos “Amazon S3”



Hacemos click en el nodo y elegimos “Change Schema” en Node Parents



Name

Amazon S3

Node parents

Choose which nodes will provide inputs for this one.

Choose one or more parent node

Q Filter parent nodes

☐ Data sources

☐ Amazon S3
S3 - DataSource

☒ Transforms

☒ Change Schema
ApplyMapping - Transform

☐ Unclassified nodes

Elegimos "Parquet" como formato

Format

Parquet

After you save your job, it will use Glue Studio's optimized Parquet writer.

En target bucket elegimos el bucket en el que queremos guardar el trabajo
y añadimos después la carpeta correspondiente

Choose an S3 bucket

S3 buckets

Buckets (1/5)

Q Find bucket

Name
☐ aws-logs-777558146599-us-east-1
☒ clima-bucket-pr
☐ consultasclima
☐ demo-bucket-pff
☐ hdf-bucket

Cancel Choose

S3 Target Location

Choose an S3 location in the format `s3://bucket/prefix/object/` with a trailing slash (/).

🔍 `s3://clima-bucket-pr/parquet/` ✕

View ↗

Browse S3

Modificamos el nombre y guardamos el trabajo que hemos creado

job_espana ✎

Visual

Script

Job details



⚠ Job has not been saved

Actions ▼

Save

Run

Your jobs (1) Info



Actions ▼

Run job

< 1 > ⚙

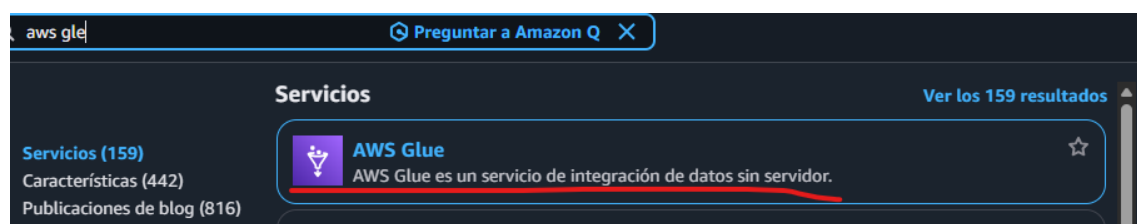
🔍 Filter jobs by property

☐	Job name	Type	Created by	Last modified	AWS Glue version	Action
☐	job_espana	Glue ETL	Visual	19/1/2026, 9:45:53	5.0	-

APARTADO C

1.- Crea un Crawler AWS GLUE que nos explore el bucket del ejercicio anterior (parquet) generando la tabla correspondiente en la base de datos clima. Ponle de prefijo a la tabla esparq_.

Seguimos los mismos pasos del ejercicio 4 y creamos el crawler



Zero-ETL integrations [new](#)

▼ Data Catalog

Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

► Data Integration and ETL

Last updated (UTC)
January 19, 2026 at 08:54:05



Action ▼

Run

Create crawler

Set crawler properties

Crawler details [Info](#)

Name

Name can be up to 255 characters long. Some character set including control characters are prohibited.

Description - *optional*

Descriptions can be up to 2048 characters long.

► Tags - *optional*

Use tags to organize and identify your resources.

[Cancel](#)

[Next](#)

Add data source



Data source

Choose the source of data to be crawled.

S3



Network connection - *optional*

Optionally include a Network connection to use with this S3 target. Note that each crawler is limited to one Network connection so any other S3 targets will also use the same connection (or none, if left blank).



Clear selection

Add new connection

Location of S3 data

- ☒ In this account
- ☐ In a different account

S3 path

Browse for or enter an existing S3 path.

s3://clima-bucket-pr/parquet/



View

Browse S3

All folders and files contained in the S3 path are crawled. For example, type s3://MyBucket/MyFolder/ to crawl all objects in MyFolder within MyBucket.

Subsequent crawler runs

This field is a global field that affects all S3 data sources.

- ☒ Crawl all sub-folders
Crawl all folders again with every subsequent crawl.
- ☐ Crawl new sub-folders only
Only Amazon S3 folders that were added since the last crawl will be crawled. If the schemas are compatible, new partitions will be added to existing tables.
- ☐ Crawl based on events
Rely on Amazon S3 events to control what folders to crawl.

☐ Sample only a subset of files

☐ Exclude files matching pattern

Cancel

Add an S3 data source

Choose data sources and classifiers

Data source configuration

Is your data already mapped to Glue tables?

☒ Not yet

Select one or more data sources to be crawled.

☐ Yes

Select existing tables from your Glue Data Catalog.

Data sources (1) [Info](#)

[Edit](#)

[Remove](#)

[Add a data source](#)

The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
☒ S3	s3://clima-bucket-pr/parquet/	Recrawl all

► Custom classifiers - *optional*

A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.

[Cancel](#)

[Previous](#)

[Next](#)

Configure security settings

IAM role [Info](#)

Existing IAM role

LabRole



[View](#)

[Create new IAM role](#)

[Update chosen IAM role](#)

Only IAM roles created by the AWS Glue console and have the prefix "AWSGlueServiceRole-" can be updated.

Lake Formation configuration - *optional*

Allow the crawler to use Lake Formation credentials for crawling the data source. [Learn more.](#)

☐ Use Lake Formation credentials for crawling S3 data source

Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, you must provide the registered account ID. Otherwise, the crawler will crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.

► Security configuration - *optional*

Enable at-rest encryption with a security configuration.

[Cancel](#)

[Previous](#)

[Next](#)

Set output and scheduling

Output configuration [Info](#)

Target database

espana



[Clear selection](#)

[Add database](#)

Table name prefix - *optional*

espparq_

Maximum table threshold - *optional*

This field sets the maximum number of tables the crawler is allowed to generate. In the event that this number is surpassed, the crawl will fail with an error. If not set, the crawler will automatically generate the number of tables depending on the data schema.

Type a number greater than 0

► Advanced options

Crawler schedule

You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like [cron](#) syntax. [Learn more.](#)

Frequency

On demand

[Cancel](#)

[Previous](#)

[Next](#)

2.- Guarda el rastreador, pero no lo ejecutes.

Al final de los pasos, hacemos click en “Crear Crawler”

Review and create

Step 1: Set crawler properties Edit

Set crawler properties

Name crawler_parquet	Description -	Tags -
--------------------------------	-------------------------	------------------

Step 2: Choose data sources and classifiers Edit

Data sources (1) [Info](#)
The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
S3	s3://clima-bucket-pr/parquet/	Recrawl all

Step 3: Configure security settings Edit

Configure security settings

IAM role LabRole	Security configuration -	Lake Formation configuration -
----------------------------	------------------------------------	--

Step 4: Set output and scheduling Edit

Set output and scheduling

Database espana	Table prefix - optional espparq_	Maximum table threshold - optional -	Schedule On demand
---------------------------	--	--	------------------------------

Cancel Previous Create crawler

APARTADO D

1.- Crea un disparador (trigger) -puedes llamarlo espa_ab - que después de finalizado el crawler del apartado A lance el trabajo del apartado B.

Dentro de “Data catalog”, hacemos click en “Data integration and ETL” y seleccionamos “Triggers”

▼ Data Integration and ETL

Zero-ETL integrations [New](#)

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Interactive Sessions

Data classification tools

Sensitive data detection

Record Matching

Triggers

Workflows (orchestration)

Blueprints

Security configurations

Cost management [New](#)

Hacemos click en “Add trigger” y seguimos los pasos para crear el trigger

Last updated (UTC)
January 19, 2026 at 09:12:10



Action ▼

Add trigger

Set trigger properties

Trigger details

Name

espa_ab

Name can be up to 255 characters long. Some character set including control characters are prohibited.

Description - optional

Enter a description

Descriptions can be up to 2048 characters long.

Trigger type

☐ On demand

Fire the trigger immediately when started.

☐ Schedule

Fire the trigger on a timer.

☒ Job or crawler event

Fire the trigger when job or crawler events match your watched list.

Watched resources (0)

List of conditions that will start the trigger

Remove

Add a watched resource

Type	Name	Status
------	------	--------

You don't have any watched resources

Add a watched resource

Add condition

Resource type

Choose a resource type

Q |

Job

Crawler

Add condition

Resource type

Crawler

The type of resource to apply this condition to

Crawler name

crawler-espana

Crawler to apply this condition to

Status

Succeeded

The status transition that the trigger listens for

CancelAdd

Watched resources (1)

RemoveAdd a watched resource

List of conditions that will start the trigger

Type	Name	Status
Crawler	crawler-espana	Succeeded

Tags - optional

Use tags to organize and identify your resources.

CancelNext

Resources to trigger (0)

EditRemoveAdd a target resource

List of resources to start once the trigger activates

Type	Name	Parameters
------	------	------------

You don't have any target resources

Add a target resource

CancelPreviousNext

Add target

Resource type

Choose a resource type

Q

Job

Crawler

Add target



Resource type

Job

The type of resource for this trigger target

Job name

job_espana

Job to start when this trigger fires

► Parameters passed down to job "job_espana" when started - optional

Cancel

Add

Choose jobs or crawlers to activate

Resources to trigger (1)

Edit Remove Add a target resource

List of resources to start once the trigger activates

Type	Name	Parameters
Job	job_espana	-

Cancel Previous Next

Review and create

Step 1: Set trigger properties

Edit

Trigger details

Name	Description	Tags
espa_ab	-	-

Watched resources (1)

List of conditions that will start the trigger

Type	Name	Status
Crawler	crawler-espana	Succeeded

Step 2: Choose jobs or crawlers to activate

Edit

Resources to trigger (1)

List of resources to start once the trigger activates

Type	Name	Parameters
Job	job_espana	-

☐ Enable trigger on creation

Cancel Previous Create

2.- Crea un disparador (trigger) -puedes llamarlo espa_bc - que después de finalizado el trabajo del apartado B lance el trabao del apartado C.

Seguimos los pasos del anterior ejercicio, pero cambiando la condición y el objetivo

Set trigger properties

Trigger details

Name

espa_bc

Name can be up to 255 characters long. Some character set including control characters are prohibited.

Description - optional

Enter a description

Descriptions can be up to 2048 characters long.

Trigger type

☐ On demand

Fire the trigger immediately when started.

☐ Schedule

Fire the trigger on a timer.

☒ Job or crawler event

Fire the trigger when job or crawler events match your watched list.

Conditional logic

How to combine conditions to determine when the trigger fires.

☒ All

All watched crawler/job event are checked before firing.

☐ Any

Any watched crawler/job event will fire the trigger.

Watched resources (0)

List of conditions that will start the trigger

Type	Name	Status
You don't have any watched resources		

Add a watched resource

Remove

Add a watched resource

Add condition

Resource type

Job

The type of resource to apply this condition to

Job name

job_espana

Job to apply this condition to

Status

Succeeded

The status transition that the trigger listens for

Cancel

Add

Watched resources (1)

Remove

Add a watched resource

List of conditions that will start the trigger

Type	Name	Status
☒ Job	job_espana	✔ Succeeded

Tags - optional

Use tags to organize and identify your resources.

Cancel

Next

Resources to trigger (0)

Edit

Remove

Add a target resource

List of resources to start once the trigger activates

Type	Name	Parameters
You don't have any target resources		

Add a target resource

Cancel

Previous

Next

Add target

✕

Resource type

Crawler

The type of resource for this trigger target

Crawler name

crawler_parquet

Crawler to start when this trigger fires

Cancel

Add

Choose jobs or crawlers to activate

Resources to trigger (1)

Edit

Remove

Add a target resource

List of resources to start once the trigger activates

Type	Name	Parameters
☒ Crawler	crawler_parquet	-

Cancel

Previous

Next

Review and create

Step 1: Set trigger properties

Edit

Trigger details

Name	Description	Tags
espa_bc	-	-

Watched resources (1)

List of conditions that will start the trigger

Type	Name	Status
Job	job_espana	✔ Succeeded

Step 2: Choose jobs or crawlers to activate

Edit

Resources to trigger (1)

List of resources to start once the trigger activates

Type	Name	Parameters
Crawler	crawler_parquet	-

☐ Enable trigger on creation

Cancel

Previous

Create

3.- Finalmente hemos de crear un trigger bajo demanda que nos arranque el crawler inicial (en nuestro caso el del apartado A)

Seguimos los pasos de los otros dos ejercicios, pero además de cambiar el target, elegimos “On demand” como el tipo de trigger

Trigger details

Name

espa_a

Name can be up to 255 characters long. Some character set including control characters are prohibited.

Description - optional

Enter a description

Descriptions can be up to 2048 characters long.

Trigger type

☒ On demand

Fire the trigger immediately when started.

☐ Schedule

Fire the trigger on a timer.

☐ Job or crawler event

Fire the trigger when job or crawler events match your watched list.

Tags - optional

Use tags to organize and identify your resources.

Cancel

Next

Resources to trigger (0)

Edit

Remove

Add a target resource

List of resources to start once the trigger activates

Type	Name	Parameters
You don't have any target resources		
Add a target resource		

Cancel

Previous

Next

Add target

Resource type

Crawler

The type of resource for this trigger target

Crawler name

crawler-espana

Crawler to start when this trigger fires

Cancel

Add

Choose jobs or crawlers to activate

Resources to trigger (1)

Edit

Remove

Add a target resource

List of resources to start once the trigger activates

Type	Name	Parameters
☒ Crawler	crawler-espana	-

Cancel

Previous

Next

Review and create

Step 1: Set trigger properties Edit

Trigger details		
Name	Description	Tags
espa_a	-	-

Step 2: Choose jobs or crawlers to activate Edit

Resources to trigger (1)
List of resources to start once the trigger activates

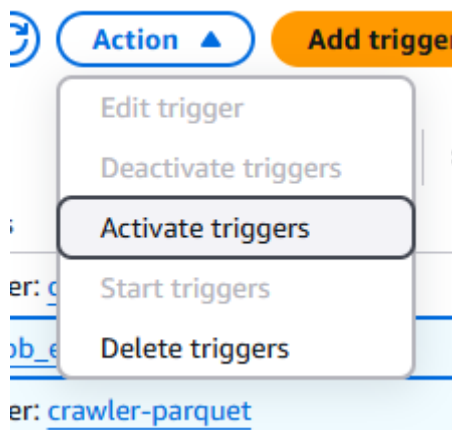
Type	Name	Parameters
Crawler	crawler-espana	-

Cancel Previous Create

4.- Arranca este manualmente este último disparador.

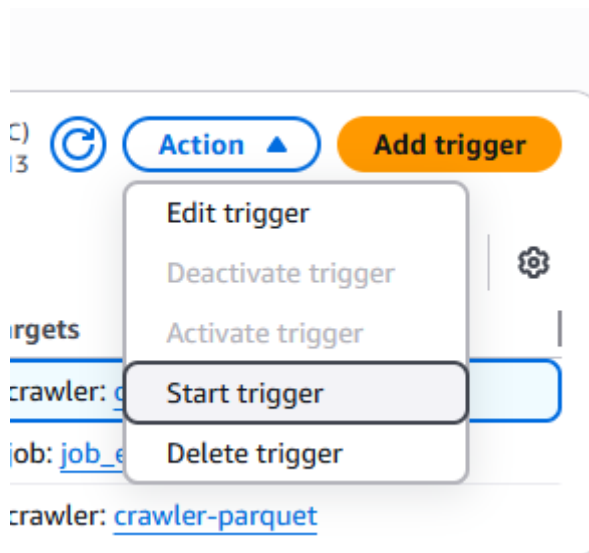
Activamos los triggers que no son “On demand” Haciendo click en la caja a su izquierda y después en Actions y Activate Triggers

☒	espa_ab	✔ Created
☒	espa_bc	✔ Created



Ahora arrancamos el trigger on demand haciendo click en la caja a a su izquierda y seleccionando Actions y Start Trigger

☒	espa_a	✔ Created
-------------------------------------	------------------------	-----------

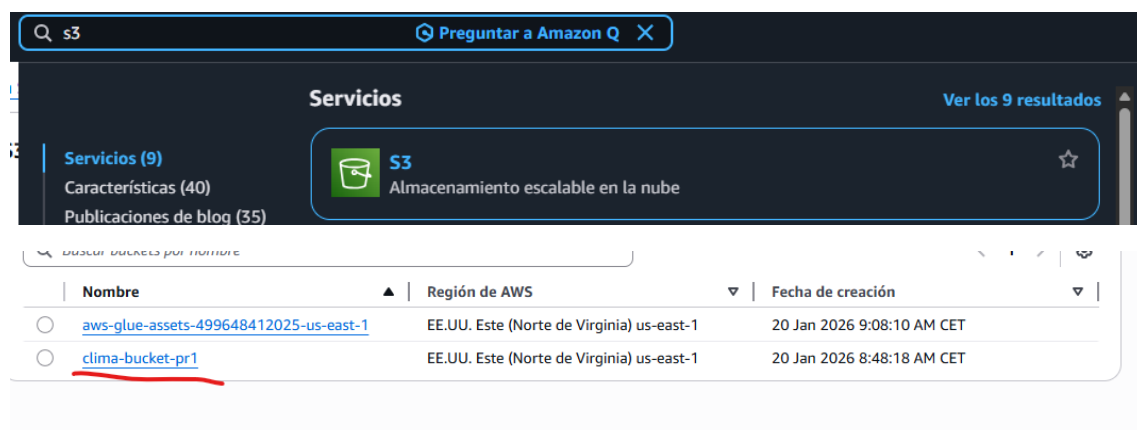


APARTADO E

- Deberían de ir ejecutándose todos los trabajos y crawlers. Cuando finalicen todas las tareas tendrían que haberse creado los archivos CSV y Parquet así como la tablas con los nombres de sus campos en español. Verifica que todo ha ido correctamente.

1.- Muestra los archivos creados.

Si vamos a S3, entramos en el bucket que hemos creado y después en la carpeta "Parquet", veremos los archivos que ha creado el trigger



Objetos (2)

[Copiar URI de S3](#)[Copiar URL](#)

Los objetos son las entidades fundamentales que se almacenan en Amazon S3. Puede utilizar el [inventario de Amazon S3](#) para que concederles permisos de forma explícita. [Más información](#)

☐	Nombre	Tipo	Última modificación
☐	espana/	Carpeta	-
☐	parquet/	Carpeta	-

Objetos (207)							
Los objetos son las entidades fundamentales que se almacenan en Amazon S3. Puede utilizar el inventario de Amazon S3 para obtener una lista de todos los objetos de su bucket. Para que otras personas obtengan acceso a sus objetos, tendrá que concederles permisos de forma explícita. Más información							
Buscar objetos por prefijo							
☐	Nombre	Tipo	Última modificación	Tamaño	Clase de almacenamiento		
☐	run-1768897988150-part-block-0-r-00000-snappy.parquet	parquet	20 Jan 2026 9:33:26 AM CET	436.0 KB	Estándar		
☐	run-1768897988150-part-block-0-r-00001-snappy.parquet	parquet	20 Jan 2026 9:33:26 AM CET	379.2 KB	Estándar		
☐	run-1768897988150-part-block-0-r-00002-snappy.parquet	parquet	20 Jan 2026 9:33:26 AM CET	433.3 KB	Estándar		
☐	run-1768897988150-part-block-0-r-00003-snappy.parquet	parquet	20 Jan 2026 9:33:26 AM CET	428.0 KB	Estándar		
☐	run-1768897988150-part-block-0-r-00004-snappy.parquet	parquet	20 Jan 2026 9:33:23 AM CET	477.6 KB	Estándar		
☐	run-1768897988150-part-block-0-r-00005-snappy.parquet	parquet	20 Jan 2026 9:33:23 AM CET	434.7 KB	Estándar		
☐	run-1768897988150-part-block-0-r-00006-snappy.parquet	parquet	20 Jan 2026 9:33:23 AM CET	427.1 KB	Estándar		
☐	run-1768897988150-part-block-0-r-00007-snappy.parquet	parquet	20 Jan 2026 9:33:23 AM CET	437.0 KB	Estándar		
☐	run-1768897988150-part-block-0-r-00008-snappy.parquet	parquet	20 Jan 2026 9:33:27 AM CET	501.4 KB	Estándar		
☐	run-1768897988150-part-block-0-r-00009-snappy.parquet	parquet	20 Jan 2026 9:33:26 AM CET	359.2 KB	Estándar		

2.- Muestra las tablas y campos creados.

Si vamos a AWS Glue, Data Catalog, databases, y hacemos click en la database creada podremos ver las tablas que han creado los crawlers

[Preguntar a Amazon Q](#)

Servicios

Ver los 159 resultados

Servicios (159)

Características (429)

Publicaciones de blog (811)

AWS Glue

AWS Glue es un servicio de integración de datos sin servidor.

Zero-ETL integrations [New](#)

▼ **Data Catalog**

Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

▼ **Data Integration and ETL**

Databases (1)

A database is a set of associated table definitions, organized into a logical g

Filter databases

☐	Name	▲	Description
☐	espana		-

Tables (2)

View and manage all available tables.

Filter tables

☐	Name	▲	Database	▼	Location	▼	Classification	▼
☐	espcsv_espana		espana		s3://clima-bucket-pr1/espa		CSV	
☐	espparq_parquet		espana		s3://clima-bucket-pr1/parq		Parquet	

espcsv_espana

Last updated (UTC) January 20, 2026 at 08:51:34 Version 0 (Current version) [Actions](#)

[Table overview](#) | [Data quality - new](#)

Table details

Name espcsv_espana	Classification CSV	Deprecated -
Database espana	Location s3://clima-bucket-pr1/espana/	Column statistics No statistics
Description -	Connection -	
Last updated January 20, 2026 at 08:51:51		

► [Advanced properties](#)

Schema

Partitions

Indexes

Column statistics - new

Schema (8)

View and manage the table schema.

Filter schemas

#	Column name	Data type	Partition key
1	id	string	-
2	date	bigint	-
3	element	string	-
4	data_value	bigint	-
5	m_flag	string	-
6	q_flag	string	-
7	s_flag	string	-
8	obs_time	string	-

espparq_parquet

Last updated (UTC)
January 20, 2026 at 08:52:11

Version 0 (Current version)

Actions

Table overview

Data quality - new

Table details

Name

espparq_parquet

Database

espana

Description

-

Last updated

January 20, 2026 at 08:35:35

Classification

Parquet

Location

s3://klima-bucket-pr1/parquet/

Connection

-

Deprecated

-

Column statistics

No statistics

Advanced properties

Schema

Partitions

Indexes

Column statistics - new

Schema (8)

View and manage the table schema.

Filter schemas

#	Column name	Data type	Partition key
1	id	string	-
2	fecha	string	-
3	elemento	string	-
4	valor_datos	string	-
5	bandera_m	string	-
6	bandera_q	string	-
7	bandera_s	string	-
8	tiempo_obs	string	-

APARTADO F

- Vete a Athena y ejecuta por duplicado (una vez sobre la tabla `espcsv_` y otra sobre la tabla `espparq_`) las mismas consultas que en la práctica anterior mostrando sus resultados y tiempos de ejecución. Obtén los tiempos obtenidos entonces y ahora sobre las dos tablas.

1.- ¿Cuántas mediciones tenemos de España?

espcsv

Consulta 1 : ✕ | Consulta 2 : ✕

1 SELECT COUNT(*) FROM "espana"."espcsv_espana" where id like 'SP%';

Tiempo en cola: 100 ms Tiempo de ejecución: 1.258 sec Datos analizados: 347.65 MB

Resultados (1)

🔍 Filas de búsqueda	
#	▼ _col0
1	10583191

Espparq

1 SELECT COUNT(*) FROM "espana"."espparq_parquet" where id like 'SP%';

Tiempo en cola: 100 ms Tiempo de ejecución: 789 ms Datos analizados: 30.34 KB

#	▼	_col0
1		10583191

2.- Sabiendo los códigos de las 4 estaciones de Asturias ¿Cuántas mediciones tenemos de Asturias?

Espcsv

1	SELECT COUNT(*) FROM "espana"."espcsv_espana" where id like 'SPE00119792'
2	OR id like 'SPE00119801'
3	OR id like 'SPE00119819'
4	OR id like 'SPE00119828';

Tiempo en cola: 120 ms Tiempo de ejecución: 1.392 sec Datos analizados: 347.65 MB

Resultados (1)

🔍 Filas de búsqueda

#	▼	_col0
1		272023

Espparq

1	SELECT COUNT(*) FROM "espana"."espparq_parquet" where id like 'SPE00119792'
2	OR id like 'SPE00119801'
3	OR id like 'SPE00119819'
4	OR id like 'SPE00119828';

Tiempo en cola: 107 ms Tiempo de ejecución: 810 ms Datos analizados: 0.64 KB

#	▼	_col0
1		272023

3.- ¿Cuántas mediciones tenemos de Oviedo?

Espcsv

```
1 SELECT COUNT(*) FROM "espana"."espcsv_espana" where id like 'SPE00119828';
```

Tiempo en cola: 106 ms Tiempo de ejecución: 1.145 sec Datos analizados: 347.65 MB

#	▼	_col0
1		73047

Espparq

```
1 SELECT COUNT(*) FROM "espana"."espparq_parquet" where id like 'SPE00119828';
```

Tiempo en cola: 100 ms Tiempo de ejecución: 885 ms Datos analizados: 0.20 KB

#	▼	_col0
1		73047

4.- ¿Cuál es la medición más antigua de España, Asturias y Oviedo?

Espcsv

```
(SELECT * FROM "espana"."espcsv_espana" where id like 'SP%' ORDER BY date LIMIT 1)
UNION
(SELECT * FROM "espana"."espcsv_espana" where id like 'SPE00119792'
OR id like 'SPE00119801'
OR id like 'SPE00119819'
OR id like 'SPE00119828' ORDER BY date LIMIT 1)
UNION ALL
(SELECT * FROM "espana"."espcsv_espana" where id like 'SPE00119828' ORDER BY date LIMIT 1);
```

Tiempo en cola: 107 ms

Tiempo de ejecución: 5.261 sec

Datos analizados: 1.02 GB

 Filas de búsqueda

#	▼	id	▼	date	▼	element	▼	data_value
1		SPE00119828		19721201		TMAX		130
2		SPE00119801		19381001		TMAX		192
3		SPE00155329		18961101		TMAX		155

Espparq

```
1 (SELECT * FROM "espana"."espparq_parquet" where id like 'SP%' ORDER BY fecha LIMIT 1)
2 UNION
3 (SELECT * FROM "espana"."espparq_parquet" where id like 'SPE00119792'
4 OR id like 'SPE00119801'
5 OR id like 'SPE00119819'
6 OR id like 'SPE00119828' ORDER BY fecha LIMIT 1)
7 UNION ALL
8 (SELECT * FROM "espana"."espparq_parquet" where id like 'SPE00119828' ORDER BY fecha LIMIT 1);
```

Tiempo en cola: 128 ms

Tiempo de ejecución: 1.118 sec

Datos analizados: 42.09 MB

Resultados (3)

 Filas de búsqueda

#	▼	id	▼	fecha	▼	elemento	▼	valor_datos	▼
1		SPE00119828		19721201		TMAX		130	
2		SPE00119801		19381001		TMAX		192	
3		SPE00155329		18961101		TMAX		155	

5.- Haz una tabla comparativa con los tiempos de ejecución de las consultas sobre las tres diferentes tablas (las de la práctica anterior y las dos de esta práctica) ¿Cuáles han sido las más veloces?

Consulta	Clima	Espcsv	Espparq
Nº España	16.501 sec	1.258 sec	789 ms
Nº Asturias	12.659 sec	1.392 sec	810 ms
Nº Oviedo	30.021 sec	1.145 sec	885 ms
Antigua España, Asturias y Oviedo	43.033 sec	5.261 sec	1.118 sec

Las más veloces han sido las de la tabla con los datos en formato parquet, debido a que es la que tiene los archivos con menor peso.